



NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Gaze Detection for Online Assessments /Driving

COURSE:

Deep Learning for Perception

INSTRUCTOR:

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GITHUB REPO :

<https://github.com/anasahmed81103/Gaze-Detection-AI>

DRIVE LINK FOR MODEL AND DATASET:

https://drive.google.com/drive/folders/151msEwr06IfVHyn4Y62qvKnpHnutfUra?usp=drive_link

1.Project Objective:

The primary objective of this project was to build a gaze detection system for use in online assessments and driving monitoring scenarios. In the context of online assessments, the tool aims to enhance academic integrity by detecting when a student diverts their gaze away from the screen—potentially indicating distraction or dishonest behavior. In driving contexts, gaze monitoring can help identify driver inattentiveness or drowsiness.

The system leverages computer vision and deep learning techniques to process webcam footage in real time and classify the direction of a user's gaze into three categories: Left, Right, and Center.

2. Datasets Used:

To build a robust and generalizable gaze detection system, we used a combination of custom collected data and publicly available datasets. These datasets helped cover a wide range of gaze directions, eye states, facial orientations, and lighting conditions

i. Custom Real-World Images Dataset

To tailor the system for realistic environments such as online assessments, a custom dataset was created. Group members captured labeled images representing different gaze directions (left, right, center) under controlled conditions. Given the limited number of images, extensive data augmentation techniques were applied, including:

- a) Rotation
- b) Zoom
- c) Contrast adjustment
- d) Brightness variation

These augmentations helped create a more generalized and sparse dataset suitable for training deep learning models.

ii. Drowsy Detection Dataset:

[Kaggle: Drowsy Detection Dataset](#)

This dataset contains labeled images of drivers in both alert and drowsy states. While not strictly a gaze dataset, it includes variations in head and eye positions that are valuable for training gaze-aware systems.

- Introduces head tilt and drowsy behavior for broader gaze detection scenarios

iii. LFR Face Dataset:

[GitHub: LFR-Face Dataset](#)

This dataset includes diverse human face images in different lighting conditions and head poses.

- Provides varied face orientations and environmental conditions
- Enhances robustness to head movements and background noise

iv. Open/Closed Eyes Dataset:

Kaggle: Open and Closed Eyes in the Wild

This dataset focuses on distinguishing between open and closed eye states, which is useful for eye detection and preprocessing before classifying gaze direction.

- Helps improve eye region extraction by detecting blinking and eye closure

v. Eye Gaze Dataset:

Kaggle: Eye Gaze Dataset

A dataset focused on gaze direction (left, right, center, up, down), featuring eye regions cropped from faces.

- Directly relevant to gaze classification
- Provided additional samples to enhance the left/right/center classification model
- Helped overcome class imbalance in custom data

vi. Challenges in Dataset Collection

Most publicly available datasets consisted of either synthetically generated 3D images or unbalanced samples, which negatively impacted training.

Datasets with only single-eye images were not compatible with the designed model, which required both eyes and facial landmarks for accurate prediction.

As a result, training with external data led to poor generalization and incorrect gaze classification.

3.Models Used:

The system was developed using a combination of traditional computer vision techniques and deep learning models to ensure both accuracy and efficiency in real-time environments.

• OpenCV Haarcascade (cv2):

We used OpenCV's pre-trained Haarcascade classifier for face and eye detection. This classical method is lightweight, fast, and effective in detecting facial features in real-time, especially under constrained environments like online assessments.

- Fast inference on CPU (no GPU required)
- Works well for frontal face detection
- Useful as a first stage to crop and isolate the eye regions for further processing by deep learning models

• **Google MediaPipe:**

We also integrated MediaPipe Face Mesh by Google to detect facial landmarks, including eye and iris positions. MediaPipe provides highly accurate 3D face tracking, making it ideal for precise gaze region extraction.

- More robust than Haarcascade in dynamic lighting and movement conditions
- Helps improve the quality of inputs to the gaze classification model by accurately locating and aligning the eyes

• **EfficientNetB0:**

The final deep learning model used for gaze classification was EfficientNetB0, a scalable and high-performing CNN developed by Google. It was fine-tuned on our custom dataset using training callbacks such as:

- ReduceLROnPlateau
- EarlyStopping
- ModelCheckpoint

These techniques helped prevent overfitting and stabilized the training process.

Why it was used:

- Pre-trained on ImageNet, offering strong feature extraction
- Lightweight and efficient — suitable for real-time applications
- Outperformed custom CNNs on our dataset in both accuracy and stability

• **Custom CNN for Gaze Classification**

A custom Convolutional Neural Network was also built and evaluated. The architecture consisted of:

- Convolutional layers for local feature extraction
- MaxPooling layers to reduce dimensionality
- Dense layers for final classification
- Softmax output with three neurons for Left, Right, and Center

Although not as performant as EfficientNetB0, this model served as a useful baseline and helped in understanding the contribution of architecture design to performance.

4. Results:

The final model yielded strong performance on the validation dataset:

Accuracy: 91%

Precision / Recall:

Left: 90% / 89%

Right: 92% / 91%

Center: 91% / 92%

Inference Speed: Real-time (approximately 25 FPS on CPU)

Confusion matrices and example predictions showed the model's ability to generalize well and highlighted occasional misclassifications under poor lighting or unusual head angles.

It demonstrated great results on test and validation data, but when tested on some new unseen images it gave a decent accuracy of upto 70% which can be further improved by more fine tuning and generalizing the dataset.

5. Discussion on Results

The system's performance validates the effectiveness of using both custom and publicly available datasets, combined with EfficientNetB0 and robust facial landmark detection through tools like MediaPipe and Haarcascade.

Strengths

- High accuracy across all three gaze categories demonstrates the model's ability to generalize well under controlled environments.
- Real-time performance on CPU confirms that the system is deployable in practical settings (e.g., during online exams or in vehicles).
- Hybrid approach of combining computer vision and deep learning allowed for better preprocessing and more meaningful features during training.

Lighting Sensitivity: Despite data augmentation, gaze classification accuracy dropped in very low-light conditions. Better lighting normalization or infrared eye-tracking could further improve results.

- **Lighting Conditions:** Prediction accuracy was sensitive to variations in lighting.

Image normalization and augmentation helped mitigate this, but further robustness is needed.

- **Head Movement:** The model performs reliably with minor head movement but tends to misclassify when users tilt their head significantly or look down.

- **User Diversity:** Differences in eye shape, skin tone, or glasses affected performance.

A more diverse training dataset would improve fairness and generalization.

- **Dataset Quality** : The current dataset, while useful, is limited in size and variety. A larger, real-world dataset from exam or driving scenarios would significantly enhance the system's performance and reliability.